

Intro to Software Engineering – Assignment 2

July 12, 2024

Assignment #2 Total: - 50 points Due Date: - 04 th Aug , 2024
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This quiz must be completed in group term project. As the assignment is based on term project, group members should be selected from term project only.

Assignment description:

In this assignment, you will explore the benefits of Continuous Integration and Continuous Deployment (CI/CD) pipelines and learn how to customize them using GitLab as the CI/CD platform and Docker for container packaging and deployment.

In this assignment, you will:

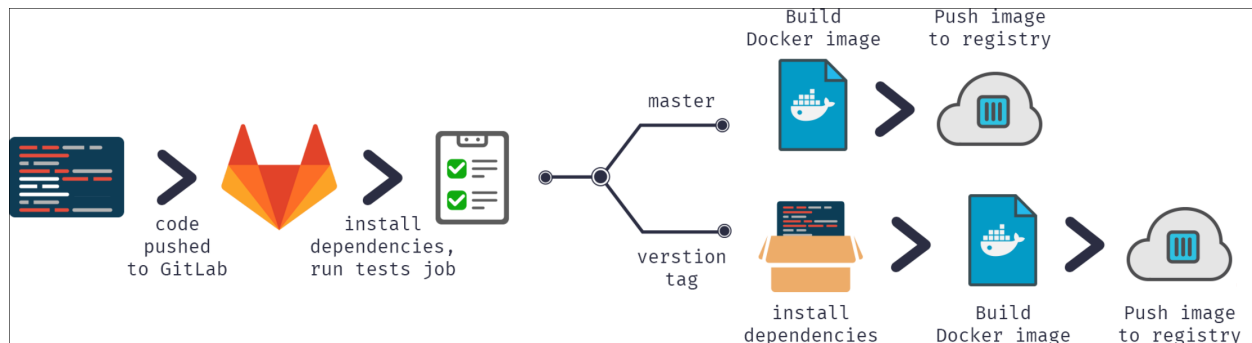
- Familiarize yourself with the concepts of CI/CD and its benefits in software development.
- Understand the role of GitLab as a CI/CD platform and explore its features and capabilities.
- Gain an understanding of Docker and its role in packaging and deploying applications.
- Customize a CI/CD pipeline using GitLab and Docker to build, test, and deploy a containerized application.
- Configure GitLab's integration with your version control system (e.g., Git) to trigger automated builds and tests whenever code changes are pushed.
- Implement stages in the CI/CD pipeline, such as building the Docker image, running tests, and deploying the container to a target environment.
- Explore the configuration options and tools available within GitLab and Docker to customize the pipeline according to your specific project requirements.
- Troubleshoot common issues that may arise during the pipeline setup and execution.
- Document the steps followed and the customizations made to the CI/CD pipeline in a report or README file.

Project description:

You have been assigned to work with the company on the project that you are working on from the beginning of this semester. The company now requires you to design a CI/CD pipeline since your company believes that this process will be essential for streamlining their software development process, enabling them to automate the build, test, and deployment stages. This automation will result in faster and more reliable delivery of their software updates. The pipeline consists of the following steps:

Part 1 (30 marks):CI Pipeline

For the CI part as figure bellow shows, you need to perform the following tasks:



CI Pipeline

1. Push your code to GitLab, including all necessary dependencies, and prepare it for running tests. (10 marks)
2. Build a Docker image for your master branch and push it to Docker Hub. (10 marks)
3. Build a Docker image that includes all required dependencies and push it to Docker Hub with an appropriate version. (10 marks)

Part 2 (20 marks): CD Pipeline

In the second part of your work, you need to build a continuous delivery pipeline that includes the following steps:

1. Pull the container from Docker Hub and deploy it on any containerized engine based on your preference. (10 marks)
2. Set up automated testing for the deployed container to ensure its functionality and performance. (10 marks)

Submission materials:

- Create a new folder in your git repository /CICD and put below material.
- Report documents, including system design and detailed explanations for each step of workflow implementation.
- Files, including configuration files, Docker file, README and automation scripts.